

UMBA, Russian Federation

WMO: 223240

Lat: 66.681N

Lon: 34.350E

Elev: 130

StdP: 14.63

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-17.2	-11.8	-21.3	1.7	-17.1	-15.2	2.4	-11.7	22.5	26.9	20.0	20.7	4.8	0	0.718

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	73.1	62.7	69.8	61.0	66.8	59.4	64.7	71.0	62.5	67.9	60.5	65.3	7.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.0	83.6	67.8	60.0	77.6	65.9	57.9	72.1	63.6	29.8	71.0	28.3	68.1	26.8	65.3	72.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.2	17.0	15.3	-22.6	78.0	7.6	3.3	-28.1	80.4	-32.6	82.3	-36.9	84.2	-42.4	86.6		
(5)			-22.9	66.7	7.6	2.4	-28.3	68.5	-32.7	69.9	-37.0	71.2	-42.5	73.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.8	14.0	13.1	21.5	31.5	41.2	51.5	58.8	55.4	47.2	36.5	25.7	19.7		
	DBStd	17.69	12.38	12.82	8.68	6.99	7.46	6.54	6.02	6.00	5.67	6.52	8.88	11.28		
	HDD50	6160	1115	1033	882	554	290	58	6	15	121	419	729	938		
	HDD65	11039	1580	1453	1347	1004	738	406	205	306	534	884	1179	1403		
	CDD50	618	0	0	0	0	17	104	279	181	37	0	0	0		
	CDD65	23	0	0	0	0	0	2	14	7	0	0	0	0		
	CDH74	70	0	0	0	0	0	9	41	20	0	0	0	0		
	CDH80	2	0	0	0	0	0	0	1	1	0	0	0	0		
	WSAvg	7.7	7.7	7.6	8.0	7.8	7.8	7.6	7.0	6.8	7.4	7.9	8.1	8.1		
	PrecAvg	19.3	1.1	0.8	0.9	1.0	1.4	1.9	2.2	2.5	2.2	2.1	1.9	1.3		
(15) Precipitation	PrecMax	27.0	2.0	2.2	1.9	2.5	3.6	5.3	4.8	7.1	5.0	4.8	4.8	2.4		
	PrecMin	13.0	0.3	0.2	0.1	0.0	0.1	0.1	0.2	0.6	0.1	0.2	0.3	0.0		
	PrecStd	3.0	0.4	0.4	0.5	0.5	0.7	1.0	1.3	1.3	1.1	1.2	0.8	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.1	37.3	41.0	52.2	66.9	74.7	78.2	76.6	63.7	51.3	43.2	38.5		
		MCWB	33.7	35.4	35.6	43.0	54.2	61.5	65.5	65.9	57.2	49.0	42.0	36.3		
	2%	DB	33.0	33.6	37.1	46.5	60.3	68.7	74.5	71.2	60.1	49.1	40.5	36.0		
		MCWB	32.3	32.5	33.6	39.6	50.2	58.4	63.8	62.5	54.7	47.2	39.4	34.7		
	5%	DB	31.1	31.7	34.7	43.0	56.4	65.1	71.3	67.5	57.6	47.2	38.0	34.2		
		MCWB	30.4	30.7	32.1	37.5	48.0	55.4	61.9	61.1	53.6	45.5	36.9	33.2		
	10%	DB	29.2	29.5	32.7	40.5	52.8	62.0	68.5	64.0	55.4	45.3	36.1	32.6		
		MCWB	28.5	28.6	30.8	35.8	45.9	53.6	60.6	58.5	52.3	43.8	35.1	31.8		
	0.4%	WB	34.1	35.7	36.5	43.4	55.4	63.7	67.8	67.4	58.0	50.0	42.2	36.8		
		MCDB	34.7	37.2	39.6	50.1	64.2	71.6	74.3	74.2	62.0	50.6	42.8	37.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	32.5	32.7	34.1	40.5	51.4	59.6	65.7	64.1	56.0	47.7	39.7	35.0		
		MCDB	32.8	33.3	36.3	45.6	58.1	66.4	72.0	70.1	59.0	48.7	40.5	35.9		
	5%	WB	30.5	30.7	32.6	38.3	48.9	56.8	63.5	61.6	54.3	45.8	37.2	33.3		
		MCDB	31.0	31.5	34.2	42.0	55.5	63.1	69.3	66.1	57.0	46.8	37.9	34.1		
	10%	WB	28.5	28.6	31.2	36.5	46.4	54.9	61.6	59.3	52.6	43.9	35.2	32.0		
		MCDB	29.1	29.5	32.8	39.6	51.9	60.5	66.9	63.2	54.9	45.2	35.9	32.6		
	5% DB	MDBR	9.7	10.2	11.7	10.8	11.0	11.6	11.2	9.9	8.6	6.3	7.1	8.6		
		MCDBR	6.9	7.6	9.6	12.7	16.8	15.8	15.5	12.9	10.7	6.8	5.7	6.2		
		MCWBR	6.9	7.2	7.8	8.2	9.3	7.8	7.3	6.5	6.7	5.5	5.4	6.0		
		MCWBR	7.3	7.6	8.6	11.6	15.4	14.0	13.4	11.8	9.2	6.2	5.4	6.2		
(40) Clear-Sky Solar Irradiance	taub		0.214	0.272	0.302	0.333	0.348	0.347	0.371	0.348	0.324	0.289	0.354	N/A		
		taud	1.938	2.067	2.129	2.215	2.389	2.460	2.445	2.484	2.522	2.311	2.615	N/A		
	Edn at Noon		54	174	234	260	270	273	261	256	236	174	61	0		
		Edh at Noon	5	17	28	33	31	30	30	26	20	14	5	0		
(44) All-Sky Solar Radiation	RadAvg		19	164	617	1156	1481	1718	1565	1086	593	198	36	0		
	RadStd		2	29	60	79	109	162	187	101	59	29	5	0		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		+0.97	N/A	N/A	N/A	N/A	N/A	-227	-320	N/A	N/A

Nomenclature: See separate page